CLASS 10 — MATHS · FORMULA SHEET

Chapter 12: **Surface Areas and Volumes** | Quick revision, one page

Cuboid and cube

- Cuboid: $V = lbh$; TSA $= 2(lb + bh + hl)$; LSA $= 2h(l + b)$; diagonal $= \sqrt{l^2 + b^2 + h^2}$.
- Cube (side a): $V = a^3$; TSA $= 6a^2$; LSA $= 4a^2$.

Cylinder (radius r, height h)

- CSA $= 2\pi rh$; TSA $= 2\pi r(r + h)$; $V = \pi r^2 h$.

Cone (radius r, height h, slant height l)

- $l = \sqrt{r^2 + h^2}$; CSA $= \pi rl$; TSA $= \pi r(l + r)$; $V = \frac{1}{3}\pi r^2 h$.

Sphere and hemisphere (radius r)

- Sphere: SA $= 4\pi r^2$; $V = \frac{4}{3}\pi r^3$.
- Hemisphere: CSA $= 2\pi r^2$; TSA $= 3\pi r^2$; $V = \frac{2}{3}\pi r^3$.

Combination of solids

- Total surface area of the combined solid = sum of the **exposed** surfaces only (joined faces are not counted).
- Volume of the combined solid = sum of the volumes of the parts.
- Typical solids: cone on a hemisphere, cylinder with hemispherical ends, cone on a cylinder, hemisphere scooped out of a cylinder (subtract its volume).

Conversion of solids

- When a solid is melted and recast, volume is conserved (surface area is not).
- Number of small solids $= \dfrac{\text{volume of big solid}}{\text{volume of one small solid}}$.
- Water flowing through a pipe: volume per unit time $= \pi r^2 \times \text{speed}$.
- Use $\pi = \dfrac{22}{7}$ unless told otherwise; keep units consistent (1 m = 100 cm; $1\ \text{m}^3 = 1000$ litres).

MASTERMIND